

CS 122A Custom Laboratory Project Report

Pico/FPGA RC Signal Converter and Dashboard

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TODO: once implemented, add the part where it receives ELRS CRSF, encodes it into PPM to send to the FPGA, and the FPGA decodes the PPM and outputs two PWM signals via two wires that control a servo and an ESC.

High-level description of the project

Our project implements a multi-stage remote-control signal converter and live dashboard, using a Raspberry Pi Pico 2W to receive ExpressLRS (ELRS) CRSF telemetry and display channel values on an FPGA-driven 4.3" RGB565 LCD. The Pico parses incoming RC channel data over UART, converts it into a format usable for visualization, and streams framebuffer updates to an iCE Sugar Pro FPGA display controller.

Elements of complexity

- Real-time parsing of high-speed ELRS CRSF packets at 420000 baud using the Raspberry Pi Pico UART.
- Integration of LVGL on the Pico to render a real-time channel monitor UI with eight channels, including dynamic bar graphs and numeric labels.
- SPI-based framebuffer output from the Pico to the FPGA ribbon cable display.
- Hardware button interrupt-driven redraw requests to redraw the screen when display artifacts occur.
- Developed a custom ExpressLRS firmware for the ESP32-S3 to receive CRSF telemetry data.

CHECK UP

User guide: explain how the user will interact with the system

1. Power the Pico and FPGA display hardware.
2. Pair a remote controller to the ESP32-S3 LoRA receiver to start receiving CRSF packets.
3. The FPGA's LCD shows eight channel bars and numerical values representing RC channel positions inputs from the remote.
4. If the display fails to render properly (i.e. random artifacts), press the GP15 button to force a redraw the screen.
5. To inspect serial telemetry, connect to the Pico USB serial console and observe UART1/GP5 status output.
6. **TODO** Not yet developed, but, the part where the PWM is output

List of hardware components used

- Raspberry Pi Pico 2W
- iCE Sugar Pro FPGA with development board
- 4.3" TFT LCD 480x272 RGB565 display with PMOD interface
- ESP32-S3 LoRA development board running custom ExpressLRS firmware
- **TODO: INSERT SPECIFIC REMOTE CONTROLLER WE USED**

List of any software libraries used

- Raspberry Pi Pico C/C++ SDK
- LVGL graphics library
- Custom SPIDisplay framebuffer driver provided by Dr. Allan Knight
- CRSFReader protocol parser
- Pico `pico_stdlib`, `hardware_spi`, `hardware_uart`, and `hardware_gpio`

List of any protocols used

- ExpressLRS CRSF for RC channel telemetry input
- UART serial communication between ESP32 and Raspberry Pi Pico to transfer ExpressLRS(ELRS) data via Crossfire Serial Protocol (CRSF)
- SPI framebuffer transfer from Pico to FPGA to display
- LVGL internal display flush callbacks for screen updates

How you met the requirements listed in the proposal

We started by building a custom firmware for the ESP32-S3 LoRA to receive ELRS CRSF data from the remote controller. We then implemented the signal conversion pipeline by receiving ELRS CRSF on the Pico UART1 interface and parsing channel data with CRSFReader. The Pico renders an LVGL-based dashboard showing eight RC channels on the FPGA-driven LCD, meeting the display and monitoring requirements. The system also supports a hardware button to request display redraws, and it reports serial monitoring status over USB serial. This satisfies the proposal's goals for real-time RC signal conversion, FPGA display integration, and user feedback.

- **TODO:** Talk about the receiving PPM on the Pico to sending that data over to FPGA, then FPGA outputs two PWM channels.

Wiring diagram for the physical hardware setup

TODO: Do exact wiring

- Pico UART1 RX -> ELRS receiver TX / ESP32-S3 UART output
- Pico SPI0 MOSI/SCLK/CS -> FPGA display controller input
- Pico GP15 -> button input with pull-up to request display redraw
- 4.3" TFT LCD powered from the FPGA board and driven by the FPGA framebuffer
- Pico USB -> host PC for power and serial logging

Design Diagram

The design consists of three main subsystems:

1. ELRS receiver input: ESP32-S3 sends CRSF data to the Pico UART1 RX pin.
2. Pico signal processor: `main.cpp` polls UART1, parses CRSF frames, updates channel values, and maintains an LVGL dashboard.
3. FPGA display output: the Pico sends framebuffer data over SPI to the iCE Sugar Pro board, which drives the 480x272 LCD.

AI usage

AI assistance was used to

Acknowledgements